GitHub review and launch readiness

25 October 2026 | Prepared for Joannah Lowe

The code has materially improved. It is not yet verified for an unattended end-to-end launch.

GitHub access is now working. This review revisits all 19 earlier findings against the actual repository, checks the later changes merged while the review was running, and executes a bounded set of isolated tests. It finds real improvements and remaining defects in contact integrity, payment reconciliation and recovery after failures.

Earlier audit findings	Current source assessment
13	The reported source path has a concrete remedy. This is not a claim that the live workflow has passed.
5	Partly addressed or still require a specific integration/guard/recovery proof: F01, F03, F14, F16 and F19.
1	A corrective database definition exists, but its live application is unverified: F07.

What was actually verified

159 existing repository test cases passed across 10 separately executed suites, with 1,028 assertions and 0 failures in the executed suites. Eleven additional focused safety-rule checks passed. Separate adversarial fixtures reproduced contact corruption, false receipt matching, polling starvation and failed warranty recovery; these are not counted as normal regression passes.

Release decision

Do not treat this review as a full launch sign-off. Correct the confirmed defects, prove one complete installer-sourced job and one complete HIVIC-sourced job in an isolated environment, then verify that the tested commit, database definitions, published Mitti templates and provider configuration are the versions being launched.

Snapshot and scope

Deep review: **b77d9f6** (PR #18 merge). Later deltas inspected through **37d1efa** (PR #21), including the 'don't know' Save my spot route and final wording adjustment. The main contact, payment and safety findings are unchanged. Tests used the b77d9f6 / 605c4e5 snapshots; the final small wording delta was inspected after testing.

No application code, database records, prices, customer communications, invoices, permissions or deployments were changed by this review. Copies and isolated fixtures were used only for analysis.

ASSESSMENT

What is better in this version

This is a substantial operational application, with meaningful business logic and safety checks. The reviewed source is more than a collection of screens. The central issue is whether all of its many paths agree on the same facts and recover correctly when a request fails or is repeated.

Shared payment calculations. The normal payment plan uses integer cents, the PIESA amount from the accepted quote, the 10% installation deposit and explicit card/bank rules. The old owner-paperwork 50/50 choice has been removed from the mounted path.

Worked example	Current calculation
Accepted quote	$3,178.00
Safety check	$349.00
Installation deposit	10% x ($3,178 - $349) = $282.90
Remaining installation balance	$2,546.10

Safety checks at shared boundaries. Booking and prestart checks use role/assignment, electrical clearance, timing, paperwork and payment facts. Missing installation evidence and unreadable evidence facts do not simply pass final approval.

Install-day facts now come from the Ready Check. The revised mapping handles no insulation, retained insulation, full removal and different area depths. It preserves a hold when required facts are absent rather than inventing them.

Failures in fixes-file retention are checked. The revised fixes path stores the certificate and completed-work evidence before advancing the PIESA record and completion state.

Reassessments and morning retries. Owner confirmation now follows the current official PIESA record. Failed day-before form creation can retry on the installation morning before work begins, with backoff and safety holds.

Real access controls. The reviewed server paths check the authenticated role, active job assignment and relevant business. The contact-update defect described next is a specific exception to that generally stronger structure.

Tests and isolation. The repository has hundreds of tests and an isolated-process runner that substitutes local test service settings. The tests are useful, but some check source text or a narrow happy path; their existence is not a complete four-role acceptance test.

Source: src/lib/payments.core.ts | src/lib/piisa-prestart.server.ts | src/lib/fixes-day.server.ts | scripts/test-isolated.ts

PRIORITY FINDINGS

Confirmed problems to fix first

R01 - Rejected installer intake changes an existing owner's contact

HIGH / reproduced | The shared enquiry helper finds an owner by email and overwrites name and phone before checking whether the address already has an open job.

Why it matters. A rejected installer submission can leave another job's shared homeowner contact changed. Correspondence or later verification can then use the wrong phone. The reproduction demonstrates contact corruption, not a completed account takeover.

Correction. Check duplicate address and authorization before contact mutation. An installer intake must not update a contact shared with another business's jobs without an explicit authorized update path. Keep the rejection atomic with respect to customer data.

Acceptance check. Installer B submits the email/address of an existing job with a different name and phone. The request is rejected, the office may receive its duplicate flag, and the original owner record remains byte-for-byte unchanged.

Source: src/lib/enquiry.server.ts | src/lib/partner-import.server.ts | src/lib/partner-import.server.ts

R02 - Two independent payments can be treated as one

HIGH / reproduced | The reconciliation fallback matches equal amounts within ten days even without a matching Stripe receipt identity. A $349 card payment plus a separate $349 bank transfer returned $349 credited, instead of $698.

Why it matters. An owner can be shown as owing money already received, held from the next step, or asked to pay again. The positive control for one receipt with its exact Stripe identity correctly credits only $349.

Correction. Match using a reliable provider receipt mapping. Treat an ambiguous same-amount/date pair as a reconciliation decision; do not silently assume that two payments are one.

Acceptance check. Separate $349 card and $349 bank payments credit $698. The same card receipt subsequently appearing in Xero credits $349 once. Repeat delivery, reversal and refund cases preserve the correct net amount.

Source: src/lib/payments.core.ts

PRIORITY FINDINGS

Recovery and access defects

R03 - Xero fallback polling can never reach newer unpaid invoices

MEDIUM / reproduced / relevant when fallback polling is relied on | The query takes the oldest 40 linked invoices before skipping settled or closed jobs. Skipped records retain their old sync timestamp. In a fixture with 40 older settled invoices and an unpaid 41st, two polls checked zero invoices.

Why it matters. A missed or disabled webhook can leave a real payment unreflected indefinitely even while the poll reports no errors. This is a source defect; its live occurrence was not established.

Correction. Filter eligible invoices before the processing limit, or page through candidates until the bounded number of eligible invoices has been processed.

Acceptance check. With 40 old settled invoices ahead of an unpaid one, a bounded run still reaches and refreshes the unpaid invoice.

Source: src/lib/xero.server.ts

R04 - Warranty follow-up can be consumed by a failed send

MEDIUM / reproduced | The new late-warranty sender records warranty_followup_sent before attempting delivery. The next call treats that marker as already sent, even if the first delivery failed and warranty_copy_provided was never recorded.

Why it matters. The original missing-PDF recovery now exists, but a failed recovery attempt can leave the owner without the warranty and the AP-signing prerequisite outstanding.

Correction. Separate reservation, confirmed delivery, definite failure and unknown outcome. Permit a safe retry after a definite failure using the same provider idempotency key; never blindly retry an unknown delivery outcome.

Acceptance check. First send fails definitively. Once the sender is healthy, a later tick sends one copy, records warranty_copy_provided and allows the AP prerequisite to be reassessed.

Source: src/lib/warranty-followup.server.ts

R05 - A second owner row using an existing mobile can fail SMS verification

MEDIUM / reproduced with a uniqueness-enforcing auth fixture. The helper reuses an existing user by verified email, but the SMS route tries to create a user again for an already registered mobile. This affects separate unlinked owner rows sharing a mobile, not every homeowner login. Resolve and link an existing identity only after verifying that mobile; test two properties and preserve account isolation.

Source: src/lib/owner-verify.server.ts

Supabase Auth error-code reference documents phone_exists for an already registered mobile; live account behavior was not tested.

BOUNDED PROOF REQUIRED

Integration questions that still block sign-off

R06 - Concurrent Stripe events need an atomic receipt claim

HIGH conditional risk. The reviewed handler reads the ledger, then inserts a payment event. In an isolated interleaving, two callbacks for one PaymentIntent both read an empty ledger, both insert and apply $698 for one $349 receipt. The fixture has no receipt-specific unique constraint. The live schema and every repository migration were not exhaustively verified, so this is not an unconditional production claim.

Required proof: a database-enforced unique receipt identity or an atomic claim/insert transaction, exercised by concurrent callbacks and later retries. Both event types must resolve to one PaymentIntent, one accounting entry and one downstream progression.

Source: src/lib/stripe.server.ts | src/lib/workflow.server.ts

R07 - Repairs must amend the same Xero invoice

HIGH integration gap in the reviewed paths. The owner repairs approval/pay/book screens are now present, but the inspected approval path does not amend the existing Xero invoice. The inspected invoice writer returns once an invoice is linked, while the portal adds the accepted repairs to its local payment plan.

For a $4,000 installation plus $550 later-approved electrical repairs, prove that the same Xero invoice contains the repairs line and totals $4,550; the portal, paid status, balance and final invoice must agree. A separate trigger or amendment writer elsewhere may exist and needs to be demonstrated. This review does not claim that every possible writer was ruled out.

Source: src/lib/portal.functions.ts | src/lib/payments.server.ts | src/lib/xero.server.ts | src/lib/xero.server.ts

R08 - Electrician-at-start exception needs a complete payment and safety test

Attendance coordination, the A5 'electrician here' answer/photo and an exception clearance path now exist. However, the clearance reader labels the result of the generic paymentGate as fixesPaymentCleared. The generic gate can clear on the safety-check payment alone. Prove that the full accepted onsite electrical charge is required at the intended boundary, and that attendance/sign-off evidence remains compulsory before work begins.

Source: src/lib/electrical-clearance.server.ts | src/lib/payment-gate.core.ts | src/lib/install-day.core.ts

R09 - Payout display and write guards still need to agree

The office due-date display now uses the same later-of-AP-and-owner-payment rule as the installer view. The reviewed Mark paid handler checks form/evidence and electrician payment prerequisites, but does not independently enforce the A1 owner-full-payment/AP timing rule for installation payouts. This action records payment; it does not itself move money. Apply the same eligibility rule to the write boundary and test direct invocation.

Source: src/lib/office-money.server.ts | src/lib/office.functions.ts | src/lib/trade-payout-gate.core.ts

Earlier audit: findings F01-F10

Addressed means the original source path has a remedy, not that production or every alternative path has been exercised. **Partial / proof needed** means the original area is improved but should remain open. **DB unverified** means source alone cannot establish deployment.

ID	Earlier issue	Status	Current result / remaining proof
F01	Same receipt counted twice	Partial	Exact receipt matching and reversals added. Equal-value independent payments still merge; concurrent Stripe identity needs an atomic constraint proof. R02 / R06.
F02	Old 50/50 owner paperwork	Addressed	Mounted choice removed; shared paymentPlan supplies 10% figures. Verify approval of the exact new wording version in the live database.
F03	Repairs approval/pay/book disconnected	Partial	Owner route and full-payment booking guard added. Same-invoice repairs amendment remains unproved. R07.
F04	Unpaid near-start install	Addressed	Earlier provisional deadline, unpaid prestart blocker and past-start office alert added. Prestart stores its payment result: separately test a later reversal or missing payment view. Live journey not executed.
F05	Missing existing-insulation PIISA facts	Addressed	Real Ready Check area facts mapped into install-day answers. Six isolated mapping cases passed; published-form parity remains a launch check.
F06	Fixes completion despite failed file retention	Addressed	Mandatory file lookup/download/upload/insert errors now hold completion before downstream record changes.
F07	Repairs reset database deadline view	DB unverified	Corrective SQL exists in pending-migrations; runtime still reads that view. Verify deployed definition against the original official PIESA date.
F08	Reassessment reused old owner confirmation	Addressed	Reader selects latest initial or reassessment and confirmation for that exact record ID.
F09	Unsent final warning still closes lead	Addressed	Only a successful day-21 result starts the seven-day close. Failure retry and message completion remain separate operational checks.
F10	Quiet-hours reminder lost overnight	Addressed	Sweep checks quiet hours before claiming the reminder; the daytime tick can select it later.

Source: src/lib/estimate-followup.core.ts | src/lib/thirty-day-rule.server.ts | docs/pending-migrations/piesa-deadline-from-official-piesa.sql

Earlier audit: findings F11-F19

ID	Earlier issue	Status	Current result / remaining proof
F11	Reassessment inherits reminder history	Addressed	Reminder history and keys are scoped to the deadline/window, allowing a new official cycle to have its own reminders.
F12	Installer offer text sent overnight	Addressed	job_offer respects quiet hours; catch-up sweep considers current pending unexpired offers and prior claims.
F13	Undeliverable outbox notification kinds	Addressed	Six identified dead enqueues removed; approved direct owner senders and sweeps are present. This does not certify all message families.
F14	Electrician-at-start exception incomplete	Partial / proof needed	Substantial attendance and A5 mapping work exists. Complete exception journey and full electrical-payment rule still need proof. R08.
F15	No install-morning retry after failure	Addressed	Same-day-before-start retry added with one-hour backoff; original holds and TEST exclusion preserved.
F16	Late warranty never sent after completion pack	Partial	Separate late-warranty sweep added, but a failed send can permanently consume its completion marker. R04.
F17	Signed-in route skips 30-day acknowledgment	Addressed	Shared server boundary enforces acknowledgment before saving windows; current-cycle handling is present.
F18	Manual office customer-payment entry	Addressed	Normal manual entry function/UI removed; residual simulation checks active qa_test jobs.
F19	Office A1 payout due date inconsistent	Partial	Read display now shares payState. Installation Mark paid write guard still lacks the matching owner-payment/AP rule. R09.

Correct form names

A1 = installer-sourced Ready Check; A2 = HIVIC-sourced Ready Check; A3 = PIESA; A4 = electrical fixes; A5 = installation day. Older audit references use the previous numbers and must be interpreted by purpose, not reported as a renumbering defect.

Current workflow reference

The original job guide contains older wording and internally conflicting sections. This review uses the actual newer decisions where identifiable, including the current 10% deposit, address-based duplicate rule, one Xero invoice and later form renumbering. It does not adopt the old guide's assumptions as legal or tax conclusions.

Source: src/lib/ready-check-offer-text.server.ts | tests/external-action-wiring.test.ts | src/lib/piisa-autosend.core.ts

EVIDENCE BOUNDARY

What was tested, and what remains untested

Validation	Observed result
Existing repository suites	159 passed; 0 failed; 10 independently executed suites. Source and mocked-fixture tests are distinguished in the run notes.
Additional safety-rule checks	11 passed. Actual extracted functions: six existing-insulation cases and five form-retry/timing cases.
Adversarial fixtures	Contact mutation before rejection; independent same-value payments; Xero polling starvation; duplicate-mobile registration. Warranty recovery result below. Concurrent Stripe fixture is conditional on database uniqueness.
Build/full default suite	Not executed. Partial authenticated source snapshot and targeted runtime checks were used; no full-checkout production build result is claimed.
Browser journeys / live integrations	Not executed. No real owner, trade, Xero, Stripe, Mitti, Registry, email or SMS actions.

Executed suite inventory

payments.core.test.ts; stripe.core.test.ts; payments-functions-readonly.test.ts (43 cases combined); warranty-followup.test.ts (7); thirty-day-rule.test.ts (21); job-offer-quiet-hours.test.ts (4); piisa-autosend.test.ts (7); spot-release.core.test.ts (29); install-day.test.ts (16): b77d9f6, unchanged in the reviewed delta. estimate-followup.test.ts (32): 605c4e5.

Warranty recovery result

Four separate failure scenarios were reproduced against the actual warranty server and completion-pack recorder: test and website paths, each returning a definite rejection or throwing before acceptance. Every failed first attempt left a sent marker without warranty_copy_provided. After the sender recovered, neither the next sweep nor an explicit retry sent anything. These four assertions-of-defect (28 assertions) are additional to the 159 passing existing cases.

GitHub evidence

At both b77d9f6 and the later 605c4e5, GitHub returned no Actions runs, no check runs and no commit statuses. The repository tree contained no GitHub workflow configuration. PR #18 reported 148 passing cases in seven selected files plus TypeScript checking; that report is distinct from independently available runner logs. This does not imply the developers ran no tests.

At 605c4e5 there are 686 checked-in test/spec files; the default non-recursive Bun runner selects 665 top-level .test.ts/.test.tsx files. Twenty .mjs tests and one qa test sit outside that pattern. These are file counts, not case counts, and do not show whether excluded files run separately.

Deployment remains unverified

A merge is not deployment evidence. The available GitHub read evidence did not establish which commit is serving the live portal. Historical smoke-test reports from earlier dates were not reused as current deployment proof. Verify the release fingerprint from the running app and the database/template versions associated with it.

Source: scripts/test-isolated.ts | package.json

FIX AND VERIFY

The shortest credible route to launch

Use a bounded release candidate and evidence from the exact version being deployed. The main work is to reconcile the remaining integration and failure cases, not to redesign the portal.

Order	Required action	Pass condition
1	Protect customer contacts	Rejected duplicate submissions leave existing owner contact data unchanged across businesses and jobs.
2	Make receipts reliable	Distinct equal payments remain distinct; one receipt repeated/concurrent is credited once; reversals reduce net receipts; unpaid invoices cannot be starved by polling.
3	Prove post-PIESA extras	Accepted repairs amend the same invoice; portal and Xero totals/balance agree; paid-in-full means the full varied total.
4	Complete failure recovery	Definite message failures can be retried safely; unknown outcomes are visible for resolution; warranty delivery and AP prerequisites recover correctly.
5	Verify deployed definitions	Correct original-PIESA deadline view, exact approved payment wording, current A1-A5 templates and question mapping, provider settings and scheduled jobs.
6	Run both complete routes	One isolated A1 job and one A2 job complete across office, owner, installer and electrician with persisted readbacks, including required messages and documents.
7	Publish only the proved candidate	Build/test record, deployed commit identity and external configuration match the approved release. A monitored small pilot may then be appropriate.

Minimum acceptance scenarios

Owner journey. Quote acceptance, correct safety fee, agreement/signatures, PIESA times, reassessment acknowledgment and same-mobile second-property access.

Repairs journey. Exact-revision approval, full payment, offered-time booking, mandatory file-transfer failure and successful retry; separately exercise electrician-at-install-start.

Installation journey. Unpaid deposit near start, original deadline after repairs, recorded area facts, missing photo/crew/safety evidence, completion and owner documents.

Office and automation. One mutually exclusive queue placement, two simultaneous callbacks, midnight-to-morning reminders, failed sends, receipt reversals and A1/A2 payout eligibility.

Operational implication

The reviewed source supports continuing development and controlled acceptance testing. It does not yet support saying the entire operation is flawless or safe to leave unattended. A constrained pilot should be a deliberate operational decision after the relevant blockers and prerequisites are proved, with clearly assigned oversight.

HANDOFF

Review provenance and implementation notes

Exact versions

Earlier audit source: a4806b70c1215ab496325704e2e6151e307cc891.
Deep-review snapshot: b77d9f655610fca17221c7c81fbf1dae9b4ef43a.
First later delta: 605c4e541cd11ebc235ca9a2d0128a43201390b3.
Final inspected head: 37d1efab54147f252d4a6cc38a01d2964b12925a.

After testing, three further commits across four files extended truthful unknown-insulation wording to the questions/received-media emails and recorded the decision to leave the spot-saved text off. This small delta was inspected; it was not retested. It does not change R01-R09.

What the later delta changed

The 'don't know' website answer now routes to Save my spot with honest unknown-insulation wording and an owner-declared queue qualification, rather than stating insulation is present. The unknown declaration is kept separate from actually answering the older insulation questions. The calculator registration copy is spot-only. The inspected delta does not change the financial or safety handlers supporting R01-R09.

Code flags and database settings must be checked from their current values. Several checked-in notes still contain older 'held until approval' statements even after later approval comments. This review does not infer current live sending state from stale prose.

Source: src/lib/enquiry-groups.core.ts | src/lib/insulation-review.core.ts | src/components/portal/register-interest.tsx

Evidence quality

Reproductions execute the actual functions with isolated inputs and mocked I/O. They demonstrate the stated source behavior under those conditions, not a confirmed production incident. Findings link to pinned source paths.

The prior 32-page audit and the complete 'How a HIV Job Runs' guide were read. Their source defects were reassessed rather than copied forward. The current repository's documentation and tests were treated as evidence to verify, not as a substitute for execution.

Implementation boundaries

Keep pricing and approved customer wording unchanged while fixing these defects. Receipt matching, authorization, atomicity, failure-state handling and invoice variation should be corrected at the shared server/database boundaries. If a database migration is required, present its exact change and test evidence for review before applying it.

Do not remove safety gates merely to make a journey pass. A safe refusal with the correct explanation is different from a completed workflow. Test labels, screenshots, merged code and successful builds are each useful evidence, but none alone proves a complete live four-role journey.

Practical assessment

The strongest parts are the growing shared domain rules, actual role checks, evidence gates and isolated tests. The weakest parts are consistency across multiple intake/payment/message paths, receipt identity, partial writes and recovery after failures. Correcting those focused areas is more useful than replacing the entire application.